CrossMatch: Cross-View Matching for Semi-Supervised Remote Sensing Image Segmentation

Ruizhong Liu, Tingzhang Luo, Shaoguang Huang, *Member, IEEE*, Yuwei Wu, Zhen Jiang, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*— Recently, weak-to-strong consistency-based methods have yielded a remarkable performance for remote sensing image segmentation. However, they are designed within a single view, which encounters the problems of unreliable pseudo-label supervision and insufficient ability to capture informative features for the segmentation of complex remote sensing data. In this article, we propose a cross-view weak-to-strong consistency-based method, which aggregates rich information from two irrelevant views. We employ two subnets for the two views to generate view-specific features while encouraging them to yield the same prediction. Within each view, we enhance the perturbation space of data at the image level and feature level for a more robust representation. To leverage the information from unlabeled data, we propose a cross-view weak-to-strong consistency scheme, which employs the pseudo-label of the weakly augmented data in one view to supervise the model training in another view, facilitating an effective information exchange across views. To avoid identical information extraction from the two views, we propose a cross-view contrastive loss to maximize the dissimilarity of the feature representations across views, which ensures that the learned complementary information from one view provides additional helpful information for the model training in another view. Finally, we propose a cross-view discrepancy-based supervised constraint by imposing a larger weight on the areas that exist discrepant predictions across views in the cross-entropy loss, allowing the model to focus more on the hard-to-classify regions of the images. Extensive experimental results on several benchmark datasets demonstrate that our method outperforms the state-of-the-art.**

***Index Terms*— Deep learning, image segmentation, remote sensing, semi-supervised learning (SSL).**

I. Introduction

SEMANTIC segmentation, which assigns a label to each pixel, is a fundamental task in the interpretation of remote sensing images, providing detailed information of landcovers. Thus, it finds many applications in land cover mapping [1], [2], urban planning [3], [4] Earth observation [5], [6], [7], and environmental monitoring [8], [9], [10].

Deep learning has become the main technique for the segmentation of remote sensing images, achieving remarkable performance in the field. Supervised models learn an implicit nonlinear mapping function for prediction by using plenty of labeled data and often yield superior performance. However, data labeling is often labor-intensive and time-consuming, making the labeled data scarce [11], [12], [13], [14]. The limited training data significantly degrades the performance of the supervised models [15], [16], [17], [18]. Semi-supervised methods employ both the labeled data and unlabeled data for model learning, which can greatly reduce the demand for the labeled data [19], [20], [21], and thus have attracted extensive attention in recent years.

Semi-supervised semantic segmentation (SSSS) methods based on deep learning can be categorized into two classes: self-training and consistency regularization. The key issue of the semi-supervised learning (SSL) methods lies on how to fully exploit the large amount of unlabeled data to support the model training with limited labeled data [22], [23], [24]. In particular, self-training-based methods [25], [26], [27], [28], [29] take the predictions of the model itself for unlabeled images as pseudo-labels and utilize them to retrain the model in conjunction with labeled images, where thresholds are required to select reliable supervision signals for training to reduce wrong guidance. Consistency regularization-based methods [30], [31], [32], [33], [34] generally promote different perturbations of a data point to yield identical prediction, where the perturbations can be performed by data augmentations in the source data domain or by neural networks in the feature domain. This helps the model to learn more robust and diverse feature representations. However, both self-training and consistency regularization methods have their respective issues. When the diversity of unlabeled data is insufficient, the self-training-based methods might overfit certain specific samples, showing a degraded generalization ability. On the other hand, consistency regularization-based methods usually encounter the problem of unstable training, especially at the earlier stages when the model's predictions on unlabeled data are inaccurate, generating a large number of noisy supervision signals.

Received 29 August 2024; revised 31 October 2024; accepted 21 November 2024. Date of publication 27 November 2024; date of current version 5 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42301425 and Grant 42071322; in part by the "China University of Geosciences (CUG) Scholar" Scientific Research Funds at CUG (Wuhan) under Grant 2022164; in part by the Fundamental Research Funds for the Central Universities, CUG (Wuhan) under Grant CUG240628; and in part by China Postdoctoral Science Foundation under Grant 2023M743299. *(Corresponding author: Shaoguang Huang.)*

Ruizhong Liu, Tingzhang Luo, Shaoguang Huang, and Hongyan Zhang are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: rzliu@cug.edu.cn; luotz.gm@gmail.com; huangshaoguang@cug.edu.cn; zhanghongyan@cug.edu.cn).

Yuwei Wu and Zhen Jiang are with Jiangxi Institute of Land Space Survey and Planning, Nanchang 33000, China (e-mail: 94032233@qq.com; 81971056@qq.com).

Digital Object Identifier 10.1109/TGRS.2024.3507050

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

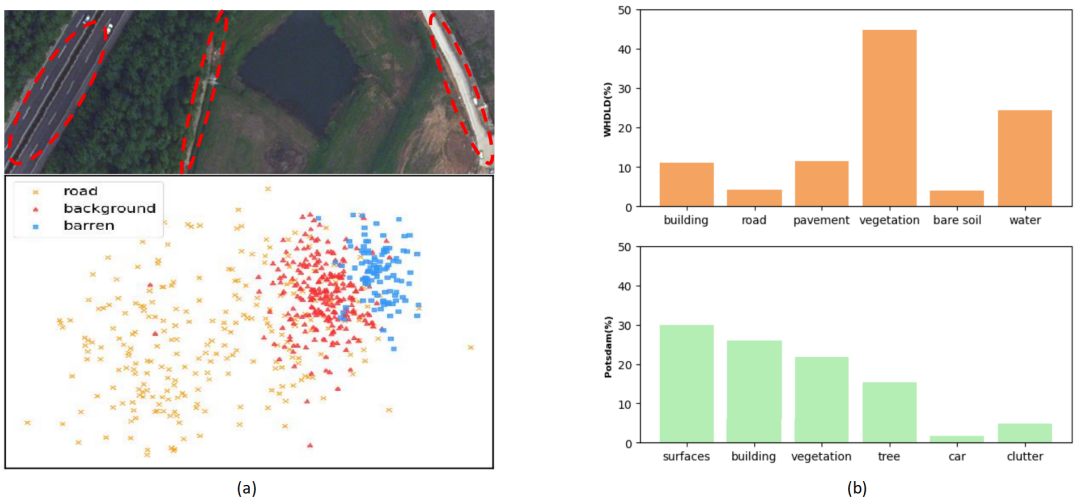


Fig. 1. (a) Scatter of data points in remote sensing image and (b) distribution (in percentage) of the classes in different remote sensing datasets.

Recent works make attempts toward integrating self-training and consistency regularization into a single framework to leverage the strengths of each scheme. Representative methods include FixMatch [35] and UniMatch [36], where a weak-to-strong consistency paradigm is employed. They typically use the predictions of the weakly augmented unlabeled samples as pseudo-labels to supervise the predictions of the strongly augmented samples within each update. These methods benefit from the supervision information acquired from weakly augmented samples and the additional information obtained from strongly augmented samples due to the expanded unlabeled data distribution, which shifts the model's decision boundary toward low-density regions and allows for more discriminative feature extraction in segmentation. It should be noted that most of these methods are designed and verified for the segmentation of natural images and related research for remote sensing images is very limited. We observe that when applying them in the segmentation of remote sensing images, we often obtain unsatisfactory results. Compared to natural images, land-cover objects of the same class in remote sensing images often exhibit larger within-class variations due to the varying structures, textures and scales, as evidenced by Fig. 1(a). This larger intraclass variation results in more false pseudo-labels during model training, significantly degrading the performance. Additionally, the interclass distribution in remote sensing images is highly imbalanced, as shown in Fig. 1(b), causing the model to be biased toward the majority classes and yield more erroneous predictions for the minority classes, further decreasing the performance. Two recent works [37], [38] based on the weak-to-strong paradigm are proposed for the semi-supervised segmentation of remote sensing images. By cross-sample image generation for improved perturbation [37] and adaptively setting the threshold for reliable pseudo-label [38], they obtained improved segmentation results. However, they are all designed within a single view and the ability to extract informative information is limited due to the hand-crafted weak and strong augmentations, which makes the models prone to yield false pseudo-labels especially when dealing with complex remote sensing data. In addition, they typically need to perform thresholding to obtain high-confidence pseudo-labels. However, a suitable threshold working for different datasets is nontrivial in practice. Both problems hinder the effective exploitation of unlabeled data.

To alleviate these issues, we propose a novel cross-view matching (CrossMatch)-based semi-supervised method for remote sensing image segmentation in this article, which can effectively use the labeled and unlabeled data. We employ two subnets to generate view-specific features while encouraging them to yield the same predictions, which allows our model to capture more informative features for segmentation from different views. To make full use of the unlabeled data, we propose a cross-view weak-to-strong consistency scheme where both the weakly and strongly augmented data are fed to each view to enhance the diversity of the data. Unlike the single-view weak-to-strong consistency methods [35], [36], [37], [38], [39], we utilize the prediction of the weakly augmented data in one view as pseudo-labels to supervise the predictions of the strongly augmented data and the feature perturbation in another view, which effectively exchange the complementary information from different views. This can refine pseudo-labeling and significantly reduce confirmation bias without setting thresholds. Meanwhile, to avoid identical feature representation from the two views, we propose a cross-view contrastive learning-based loss to maximize the representation differences between the two views for preventing networks collapse, which is beneficial for providing valuable information from one view to another. The resulting model allows us to learn informative features for classification from the two irrelevant views. To make use of labeled data, we develop a view-discrepancy-based constraint by imposing a larger weight on the pixels that exist discrepant predictions across views. This makes the model focus more on the hard-to-identify regions, facilitating a stable performance during the model training.

Our main contributions can be summarized as follows.

1) We introduce a cross-view weak-to-strong consistency scheme to gather informative knowledge from two independent views, which greatly enhances the quality of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

pseudo-labels and results in an effective usage of the unlabeled data.
2) We propose a cross-view contrastive loss to maximize the feature dissimilarity across views. This works jointly with the consistency scheme, facilitating the complementary information extraction of the input data.
3) We propose a view-discrepancy constraint for labeled data by imposing a higher penalty on the regions with discrepant predictions across views, which allows our model to focus more on the hard-to-identify areas.
4) Extensive experiments on the WHDLD, Potsdam, and LoveDA benchmark datasets show that CrossMatch achieves remarkable improvements over recent state-of-the-art.

The rest of this article is organized as follows. Section II reviews the related semantic segmentation methods. Section III presents the proposed semi-supervised segmentation method CrossMatch. In Section IV, extensive experiments and ablation studies on benchmark datasets are conducted. Finally, Section V concludes the article.

II. Related Work

In this section, we briefly review the research on the supervised semantic segmentation and the SSSS methods.

A. Supervised Semantic Segmentation

Semantic segmentation, as a dense prediction vision task, has been attracting tremendous attention [40]. Deep learning-based approaches have achieved significant achievements in the field. Among them, convolutional neural network is widely used in the supervised semantic segmentation methods. For instance, Long et al. [17] propose an encoder–decoder architecture by using a fully convolutional network for image segmentation for the first time. Inspired by this work, different extensions are obtained [41], [42], [43], [44], [45], achieving improved performance. Ronneberger et al. [41] propose a U-shape segmentation network by using skip connections to link the features of encoder and decoder. This design preserves more useful semantic information, allowing the network to achieve excellent performance with fewer samples. SegNet [42] introduces a unique upsampling strategy by utilizing pooling indices from the max-pooling layers to enhance segmentation accuracy. Chen et al. [45] develop multiple atrous convolutions at different scales during the feature extraction stage to capture multiscale information while maintaining high spatial resolution, which helps the network to better segment the multiscale objects.

Recently, transformer, a new deep neural architecture, has been adopted as the segmentation network in the semantic segmentation methods [46], [47], [48], [49]. PVT [46] is proposed by introducing a pyramid structure in the transformer-based architecture for multiscale feature extraction, improving the network's ability to model both local details and global information of the images. SegFormer [47] adopts a lightweight transformer encoder without positional encoding, avoiding the high computational complexity while maintaining a powerful ability for feature extraction. Thus, it is more scalable to large-scale images and is more well-suited to different segmentation scenarios and tasks.

B. Semi-Supervised Semantic Segmentation

SSL methods mitigate the supervised models' reliance on labeled samples by leveraging the information from a large amount of unlabeled samples. Existing SSSS methods can be grouped into two main categories: self-training-based and consistency regularization-based methods.

Self-training-based methods [25], [26], [27], [50] usually adopt a mean teacher [51] architecture, which utilizes the labeled data to train a teacher network to generate pseudo-labels of unlabeled data and retrains a student network with selected unlabeled data with high-confidence pseudo-labels and labeled data. ST++ [26] is a multistage SSSS approach based on the self-training framework, which gradually uses unlabeled images from easy to difficult and from reliable to unreliable rather than using all the unlabeled data in one update. Unlike traditional self-training-based methods that ignore unlabeled pixels with low confidence, U2PL [27] is proposed to fully exploit the potential of unlabeled data by dividing the pseudo-labels of pixels into reliable and unreliable portions and then using a pixel-wise contrastive learning method to access a more discriminative representation space for selecting accurate pseudo-labels.

Consistency regularization-based methods [30], [34], [52], [53] require the model to generate consistent predictions under different perturbations (e.g., image perturbations, feature perturbations and network perturbations). Ouali et al. [30] propose a cross-consistency training method that enforces consistency between the primary decoder and multiple auxiliary decoders, which receive different feature perturbations of the shared encoder's output as input. This approach yields more robust feature representations for the model learning. Lai et al. [52] introduce a direction-based context consistency perturbation that not only increases unlabeled data diversity but also enhances the model's sensitivity to different directions and contextual information, allowing the model to better understand complex structural objects. In addition, some attempts have been made toward the exploitation of a co-training framework, which effectively harnesses the information of unlabeled samples by using multiple networks or decoders to exchange diverse and complementary knowledge [54], [55], [56]. For instance, CPS [57] introduces cross-pseudo supervision by employing two same networks with different initializations to generate pseudo-labels to supervise and learn from each other. Fan et al. [58] propose a novel model that uses a shared backbone and two heads, which reduces the cost of both parameters and computation.

III. Methodology

In this section, we elaborate our CrossMatch method for remote sensing image segmentation. We first introduce our cross-view weak-to-strong consistency scheme. Then, a cross-view contrastive loss is developed to promote different subnets to extract complementary information across views while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

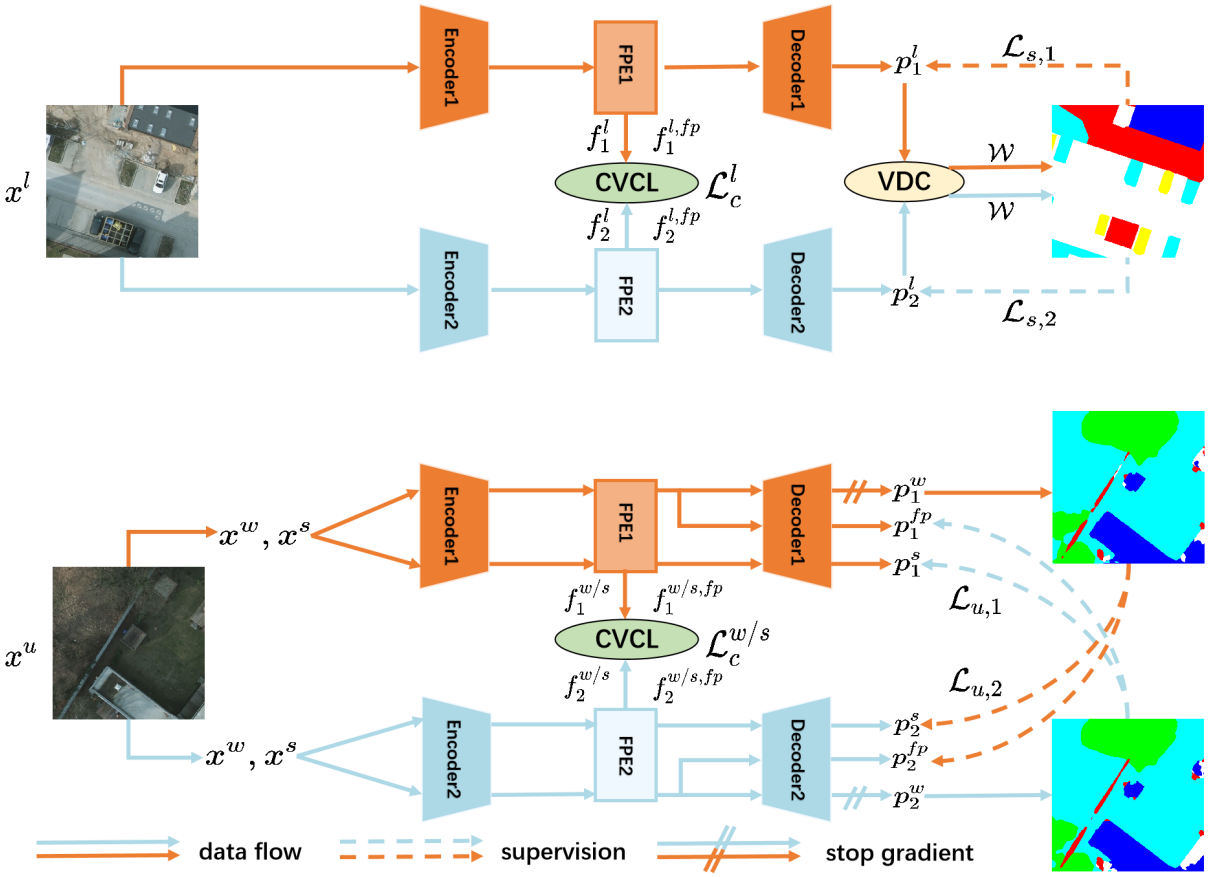


Fig. 2. Framework of the CrossMatch. We employ two unshared subnets to extract complementary information of the input data. The pseudo-labels generated in one view are used to supervise the model training in another view. $\mathcal{W}$: the weighted matrix. VDC: view-discrepancy constraint. FPE: an auxiliary feature perturbation extractor. We apply a simple feature dropout and extract both the original and perturbed features after feature fusion for the decoder. CVCL: cross-view contrastive loss.

ensuring identical predictions. Finally, we propose a view-discrepancy-based constraint for the labeled data, allowing our model to focus more on the hard-to-identify regions. The framework of the CrossMatch is shown in Fig. 2.

A. Notations

Given a few labeled images $D^l = \{(x_n^l, y_n^l) \mid x_n^l \in \mathbb{R}^{H\times W\times 3}, y_n^l \in \{0,1\}^{H\times W\times C}, 1 \le n \le M\}$ and a set of unlabeled images $D^u = \{x_n^u \mid x_n^u \in \mathbb{R}^{H\times W\times 3}, 1 \le n \le N\}$, where x_n^l/x_n^u is the nth labeled/unlabeled input image with a spatial size of $H \times W$, C is the number of classes, and M and N are the number of labeled and unlabeled images with $M \ll N$, the goal of SSSS is to train a model with the labeled and unlabeled data D^l and D^u for remote sensing image segmentation.

B. Cross-View Weak-to-Strong Consistency

The weak-to-strong consistency-based methods [35], [36], [37] employ the weakly augmented samples to obtain reliable pseudo-labels and the strongly augmented samples to expand the data distribution, allowing an effective exploitation of the unlabeled data. However, existing methods are designed in a single view where the hand-crafted augmentations limit the model's ability to obtain reliable pseudo-labels especially when dealing with remote sensing images that often have large intraclass variance and interclass imbalance. The resulting false pseudo-labels can misguide the model toward an unstable and degraded performance. In this article, we extend the single-view mode to dual views by introducing a cross-view weak-to-strong consistency strategy. The key idea is that we encourage the two views to capture different and complementary information about the scene while keeping their predictions consistent. To exchange the information between the two views, we develop a cross-view consistency loss where the prediction of the weakly augmented sample in one view is used to supervise the predictions of the strongly augmented samples and the feature-level perturbation in another view as shown in the bottom of Fig. 2.

In each view, we use an encoder–decoder architecture to perform image semantic segmentation, which is denoted by $F_i(x;\theta_i)$ $(i = 1, 2)$. The two subnets are independent and have different parameters θ_i. Specifically, a subnet F_i is divided into two parts: an encoder g_i and a decoder h_i, which are used for feature extraction and image segmentation. For each unlabeled image x_n^u, we generate its weakly augmented sample x_n^w and strongly augmented sample x_n^s by the weak augmentation $\mathcal{A}^w$ and the strong augmentation $\mathcal{A}^s$ as follows:

$$x_n^w = \mathcal{A}^w(x_n^u) \tag{1}$$

$$x_n^s = \mathcal{A}^s(x_n^w) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

where $\mathcal{A}^w$ represents the weak augmentation like random resize, flip, and rotation and $\mathcal{A}^s$ is the strong augmentation including CutMix [59] and color jitter. Both augmented samples can be seen as the new expanded samples with different levels of perturbation, which are fed to the two subnets. With the subnets $F_i(x;\theta_i)$, we can obtain the predictions of x_n^w and x_n^s, denoted by

$$p_{i,n}^w = F_i\left(x_n^w;\theta_i\right) \tag{3}$$
$$p_{i,n}^s = F_i\left(x_n^s;\theta_i\right). \tag{4}$$

Then, the pseudo-labels $\hat{y}_{i,n}^w$ of x_n^w can be obtained by applying the argmax function to $p_{i,n}^w$ for each pixel in the image.

The weak and strong augmentations are only performed at the image level for each view, which makes the student model might be difficult to yield consistent predictions against the complex structure of remote sensing data. In order to expand the perturbation space, motivated by [36] we perform perturbation in the feature space, which is denoted by the auxiliary feature perturbation extractor (FPE) in Fig. 2. Then, we obtain the raw and perturbed features as follows:

$$f_{i,n}^\alpha = g_i\left(x_n^\alpha\right) \tag{5}$$
$$f_{i,n}^{\alpha,fp} = \mathcal{P}\left(f_{i,n}^\alpha\right) \tag{6}$$

where $\alpha \in (l, w, s)$ and $\mathcal{P}$ is a perturbation operator in the feature space designed for FPE. Then, we obtain the predictions of the weakly unlabeled images with the decoder h_i for each view

$$p_{i,n}^{fp} = h_i\left(f_{i,n}^{w,fp}\right). \tag{7}$$

Basically, the two views are designed to yield view-specific features with the encoder g_i, which facilitates providing complementary information of the input data in one view for the model training in another view. The view-specific feature extraction is achieved by using a cross-view contrastive loss that is defined in Section III-C. How to exchange the information across the two views is the main problem to be solved here.

In order to leverage the information extracted from the two views, we employ the pseudo-labels generated in one view to supervise the model training in another view. As the weakly augmented samples x_n^w often yield more reliable predictions compared with the strongly augmented samples x_n^s, we use $\hat{y}_{i,n}^w$ in the ith view as pseudo-labels to supervise the predictions of the perturbed features and the strongly augmented data in another view, i.e., $p_{3-i,n}^{f_p}$ and $p_{3-i,n}^s$, as follows:

$$\mathcal{L}_u = \frac{1}{2N}\sum_{n=1}^{N}\frac{1}{2}\sum_{i=1}^{2}\frac{1}{H\times W}\sum_{m=1}^{H\times W} \times\left(\ell_{\text{ce}}\left(p_{i,n,m}^{fp},\hat{y}_{3-i,n,m}^w\right)+\ell_{\text{ce}}\left(p_{i,n,m}^s,\hat{y}_{3-i,n,m}^w\right)\right) \tag{8}$$

where ℓ_{ce} is a cross entropy loss and $p_{i,n,m}^{fp}$ represents the prediction of the mth pixel in $p_{i,n}^{fp}$. The same definition is applied to $\hat{y}_{3-i,n,m}^w$, $p_{i,n,m}^s$ and $\hat{y}_{3-i,n,m}^w$.

C. Cross-View Contrastive Loss

The success of the cross-view supervision in Section III-B lies on the assumption that the two views capture complementary information of the input data. When the two branches extract the same features, the information exchange across the two views does not provide any additional information that is valuable for segmentation. Thus, constraint is required to ensure that the two views extract view-irrelevant features, avoiding the collapse of the representations across views.

To achieve this, we propose a cross-view contrastive loss based on self-supervised contrastive learning. For each branch, the original feature $f_{i,n}^\alpha$ and its perturbed feature $f_{i,n}^{\alpha,f_p}$ are similar and should have the same prediction result. We view them as positive pairs. Meanwhile, the two branches should extract different features. Thus, we enforce the features $f_{i,n}^\alpha$ obtained in the ith view and the features $f_{3-i,n}^\alpha$ and $f_{3-i,n}^{\alpha,f_p}$ obtained in the $(3-i)$th view to be different. $f_{3-i,n}^\alpha$ and $f_{3-i,n}^{\alpha,f_p}$ are viewed as the negative samples of $f_{i,n}^\alpha$. Our cross-view contrastive loss is formulated as follows:

$$\mathcal{L}_c^\alpha = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{2}\sum_{i=1}^{2}\log \times\left(\frac{e^{\langle f_{i,n}^\alpha, f_{i,n}^{\alpha,fp}\rangle/\tau}}{e^{\langle f_{i,n}^\alpha, f_{i,n}^{\alpha,fp}\rangle/\tau}+e^{\langle f_{i,n}^\alpha, f_{3-i,n}^\alpha\rangle/\tau}+e^{\langle f_{i,n}^\alpha, f_{3-i,n}^{\alpha,fp}\rangle/\tau}}\right) \tag{9}$$

where $\langle\cdot,\cdot\rangle$ represents an inner product of two vectors, and τ is a temperature parameter. We set $\tau = 0.5$ for our model. We impose the cross-view contrastive loss on all the labeled data and the weakly and strongly augmented unlabeled data, leading to a final contrastive loss as follows:

$$\mathcal{L}_c = \mathcal{L}_c^l + \lambda_1\mathcal{L}_c^w + \lambda_2\mathcal{L}_c^s \tag{10}$$

where λ_1 and λ_2 are two parameters. We set $\lambda_1 = \lambda_2 = (1/2)$ for our model.

The proposed loss in (10) facilitates the two views to capture more diverse information of the input data. The obtained pseudo-labels from one view can bring additional information for the model training in another view, which gradually improves the quality of pseudo-labels of two views and thus finally improves overall segmentation accuracy.

D. View-Discrepancy Constraint

Supervised information is important for the model to learn discriminative features for segmentation. In order to make use of the labeled data, we encourage the two branches to yield the same segmentation result to the ground truth. The prediction results of the nth image in the ith view are obtained by

$$p_{i,n}^l = F_i\left(x_n^l;\theta_i\right). \tag{11}$$

The supervised training loss of the two views are

$$\mathcal{L}_s = \frac{1}{M}\sum_{n=1}^{M}\frac{1}{2}\sum_{i=1}^{2}\frac{1}{H\times W}\sum_{m=1}^{H\times W}\ell_{\text{ce}}\left(p_{i,n,m}^l, y_{n,m}^l\right) \tag{12}$$

where $y_{n,m}^l$ is the ground truth of the mth pixel of the nth image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

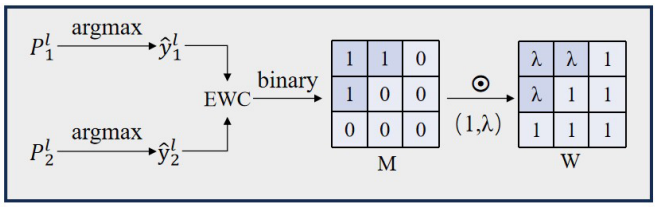


Fig. 3. Illustration of the cross-view discrepancy. The p_i^l, $\hat{y}_i^l$, $i = 1, 2$ are soft and hard labels of labeled samples from two views, respectively. By element–wise comparison (EWC) between the predictions across views, the discrepancy mask matrix $\mathcal{M}$ and the weight matrix $\mathcal{W}$ are obtained.

In fact, the supervised loss in (12) imposes the same weight, i.e., 1, on every pixel of an image, which neglects the discrepancy across the two views. For the easy-to-identify region, the two branches often can yield identical predictions. However, for some challenging regions, there might exist discrepancy for the two branches. Thus, it is more reasonable to impose larger weights on the challenging regions. Therefore, we introduce a discrepancy mask for each image to allow the model to focus more on the regions with discrepancy across views.

We first transform the soft labels of each pixel into hard labels for each image via

$$\hat{y}_{i,n}^l = \text{argmax}\left(p_{i,n}^l\right) \tag{13}$$

where the argmax operation is performed on the soft label of each pixel. Then, a binary discrepancy mask of the nth input image across views is defined by

$$\mathcal{M}_n(i,j) = \begin{cases} 1 & \text{if } \hat{y}_{1,n}^l(i,j) \neq \hat{y}_{2,n}^l(i,j) \\ 0 & \text{otherwise,} \end{cases} \tag{14}$$

where $\hat{y}_{1,n}^l(i,j)$ represents the hard label of the (i,j)th pixel of the nth image obtained in the first view. $\mathcal{M}_n(i,j) = 1$ means that there exist prediction discrepancy in the (i,j)th position. We impose a larger weight on their cross entropy loss. The weight matrix is defined by

$$\mathcal{W}_n = \lambda \odot \mathcal{M}_n + (1 - \mathcal{M}_n) \tag{15}$$

where λ is a parameter that is larger than 1 and we set it to 2 in our model. Let $\mathbf{w}_n$ represent the vectorized weight vector of $\mathcal{W}_n$, and we reformulate our supervised training loss in (12) as

$$\mathcal{L}_s = \frac{1}{M}\sum_{n=1}^{M}\frac{1}{2}\sum_{i=1}^{2}\frac{1}{W\times H}\sum_{m=1}^{W\times H} w_{n,m}\ell_{\text{ce}}\left(p_{i,n,m}^l, y_{n,m}^l\right) \tag{16}$$

where $w_{n,m}$ is the mth element of the vector $\mathbf{w}_n$.

The new training loss in (16) indicates that a larger weight λ is imposed on the regions with discrepant predictions across views as shown in Fig. 3. It is observed that the supervised loss indirectly regularizes the two views to yield identical segmentation results, which can be additionally seen as a consistency constraint across views.

E. Overall Loss

We obtain the overall loss of our model by combining the cross-view weak-to-strong loss, the cross-view contrastive loss, and the view-discrepancy based loss as follows:

$$\mathcal{L} = \mathcal{L}_u + \beta\mathcal{L}_c + \mathcal{L}_s \tag{17}$$

where β is a parameter. We set $\beta = 1/2$ for our model.

The first loss is designed for the information exchange across views by mutually supervising the training of each view. The second loss promotes view-specific feature extraction to avoid repeated information extraction. The last one is a supervised loss to make use of the valuable supervised information for discriminative feature extraction, allowing increased attention of the model on the challenging region during the model training. Unlike existing semi-supervised approaches, our model is parameter-free regarding the generation of pseudo-labels, which does not need a predefined threshold. This makes our method more flexible in real applications. Moreover, our model leverages the rich information extracted from different views, allowing our model to capture more relevant information for image segmentation compared with the existing single-view methods.

TABLE I
DETAILS OF THE BENCHMARK DATASETS

Datasets	WHDLD	Potsdam	LoveDA
Bands	RGB	RGB	RGB
Spatial resolution	2 m/pixel	0.05 m/pixel	0.3 m/pixel
Cities	Wuhan	Potsdam	Nanjing Wuhan Changzhou
The number of classes	6	6	7

IV. EXPERIMENTS

A. Datasets

We validate the performance of our model on three benchmark datasets including WHDLD [60], Potsdam [61], and LoveDA [62].

WHDLD [60] is acquired from the GaoFen-1 and ZiYuan-3 satellites, which contains six categories: building, road, pavement, vegetation, bare soil, and water. Each image has a spatial size of 256×256. The spatial resolution is 2 m/pixel.

Potsdam [61] is captured by an UltraCamXp large-format digital aerial camera. There are six categories, including impervious surface, building, low vegetation, tree, cars, and clutter. The size of the images is $6000 \times 6000 \times 3$. In our experiments, we crop each image into $512 \times 512 \times 3$ to reduce the computational load.

LoveDA [62] is a remote sensing data, covering the scene of urban areas and the rural areas of different cities. The image size is $1024 \times 1024 \times 3$. The dataset contains seven classes, including background, building, road, water, barren, forest, and agricultural. More information can be found in Table I.

B. Evaluation Metric

We evaluate the segmentation performance by using the mean intersection over union (mIoU). IoU is calculated by the ratio of the intersection area to the area of union between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

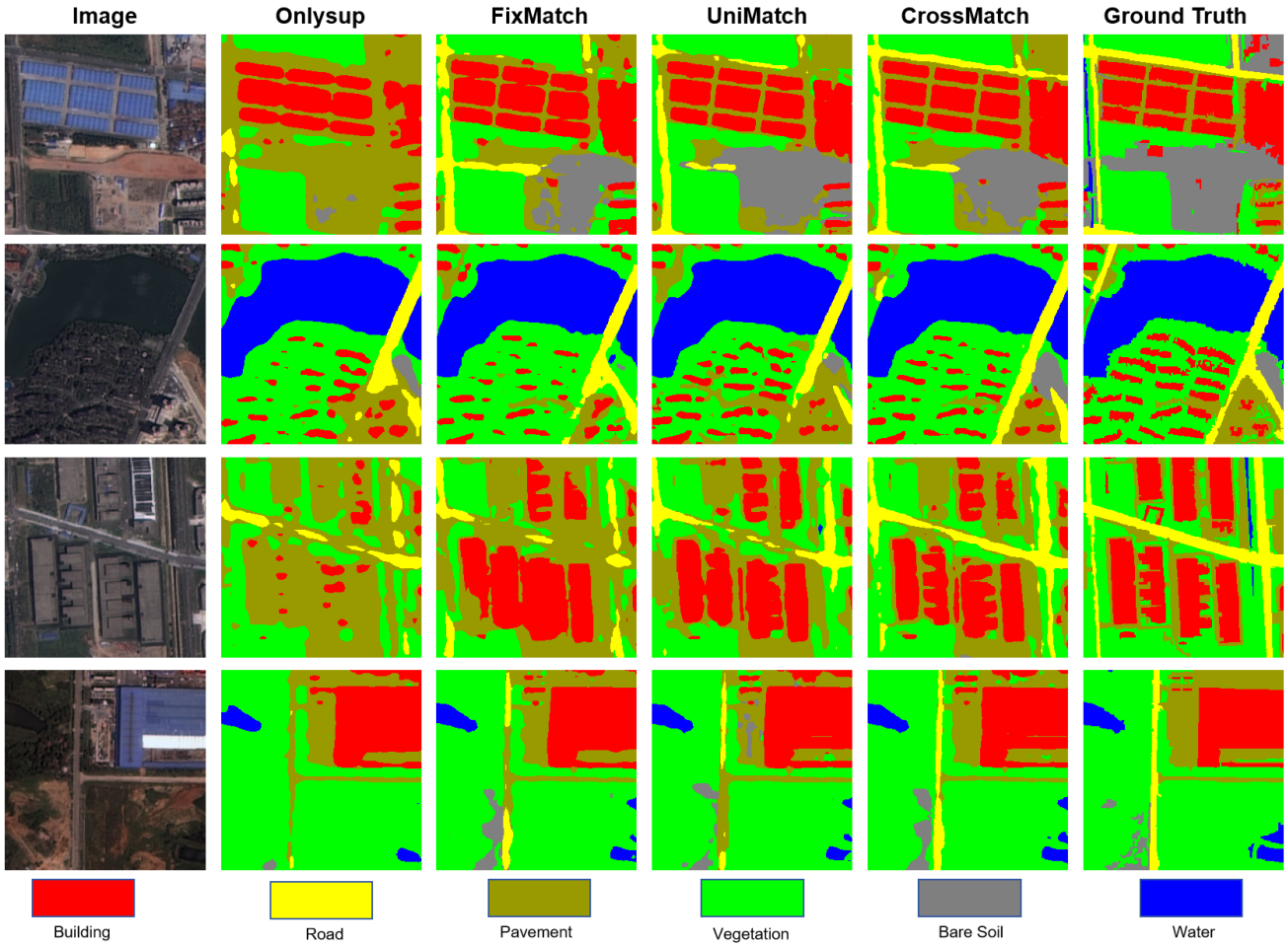


Fig. 4. Visual results of our proposed method compare with only-supervised baseline and other advanced semi-supervised methods on the WHDLD dataset.

the prediction and the ground truth. The mIoU is the average IoU across all the categories, which is calculated by

$$\text{mIoU} = \frac{1}{C}\sum_{j=1}^{C}\frac{TP_j}{TP_j + FP_j + FN_j} \tag{18}$$

where C is the number of categories and TP_j, FP_j, and FN_j are the numbers of true positives, false positives, and false negatives for the jth class, respectively.

C. Compared Methods

We compare our method CrossMatch with the recent state-of-the-art SSSS methods, including FixMatch [35], U2PL [27], WSCL [37], CCVC [63], UniMatch [36], and CorrMatch [39]. A supervised baseline Onlysup is also compared.

1) *Onlysup*: This supervised baseline uses a single network, with ResNet-50 as the encoder and DeepLabV3+ as the decoder. Only labeled samples are used during the training process.
2) *FixMatch* [35]: It is the first to propose the weak-to-strong consistency paradigm, which utilizes pseudo-labels generated from weakly augmented images to supervise strongly augmented images and enforce them to yield consistent predictions.
3) *U2PL* [27]: This method adopts a mean teacher architecture based on self-training. It first divides pixels into reliable and unreliable parts and then uses a contrastive learning method to improve the quality of the pseudo-labels.
4) *WSCL* [37]: This work is based on weak-to-strong consistency and proposes a novel cross-sample image generation method to make stronger perturbations on unlabeled data. In addition, an entropy map-based adaptive reweighting technique is designed to remove noisy pseudo-labels.
5) *CCVC* [63]: This method is built on a co-training framework, which adopts a discrepancy loss to extract distinct features from different views and proposes a conflict-based pseudo-labeling method to improve the stability of the model training.
6) *UniMatch* [36]: It is the first approach that applies weak-to-strong consistency strategy to the SSSS task. UniMatch builds on FixMatch by exploring a broader perturbation space, integrating image-level and feature-level perturbations, and also extending strong perturbations into a dual-stream.
7) *CorrMatch* [39]: This work introduces two label propagation strategies to enhance the efficiency of unlabeled data: pixel propagation, which spreads

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

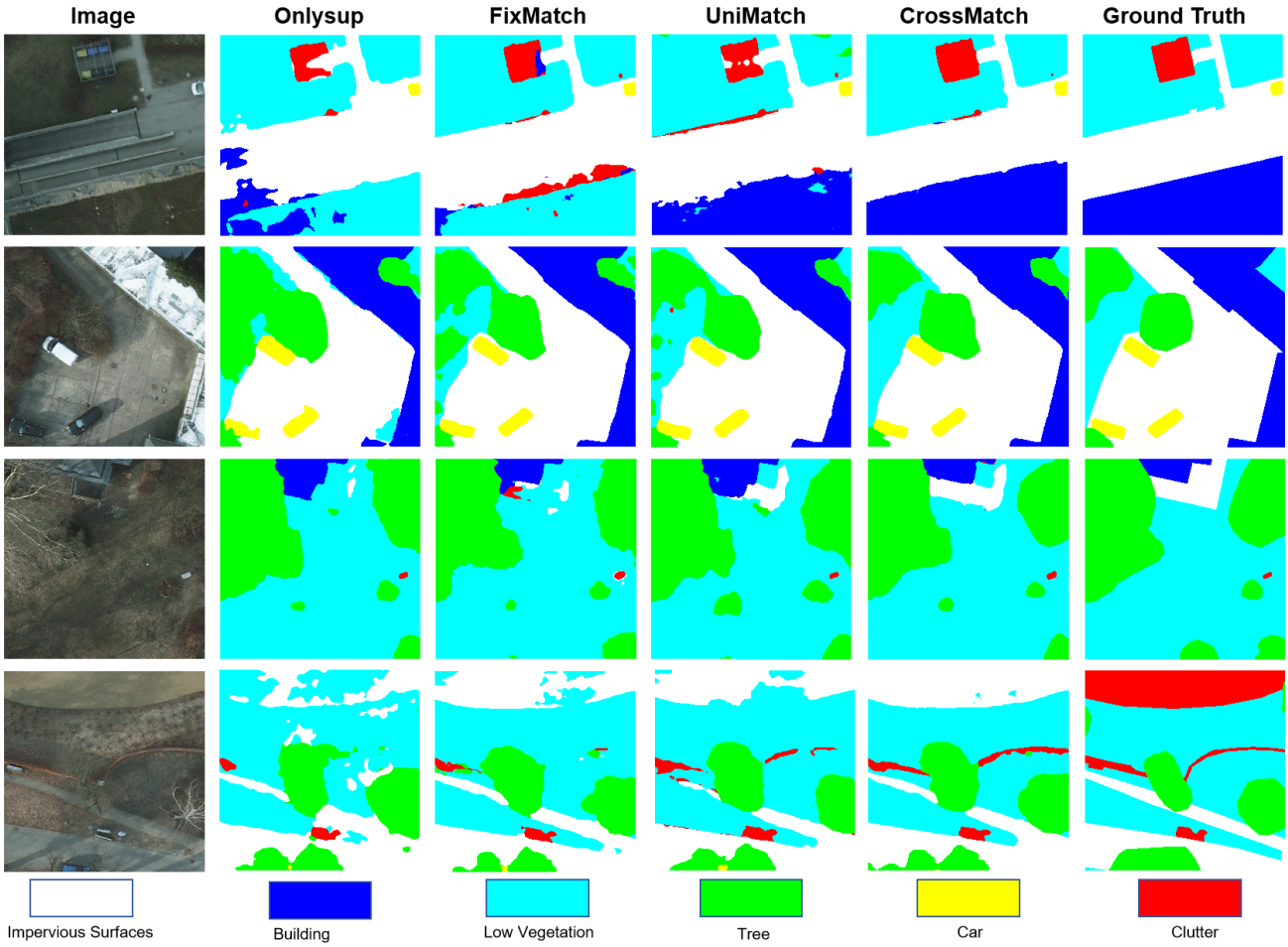


Fig. 5. Visual results of our proposed method compare with only-supervised baseline and other advanced semi-supervised methods on the Potsdam dataset.

high-confidence pixel labels based on pairwise similarities, and region propagation, which refines pseudo labels using class-agnostic masks derived from correlation maps.

D. Implementation Detail

For a fair comparison, we adopt the same backbone for all methods, where DeepLabV3+ [45] is used as the decoder and ResNet-50 [64] is used as the encoder. And we take an output stride of 16 by default. We use the stochastic gradient descent (SGD) as an optimizer. We set the base learning rate to 0.01, the weight decay to 0.0001, and the momentum to 0.9. The number of epochs for training our model is set to 80. We use the poly scheduling to decay the learning rate: $\text{lr} = \text{lr}_{\text{base}} \times (1 - (\text{iter}/\text{max_iter}))^{0.9}$. We conduct experiments in the WHDLD and Potsdam datasets on a computer with two NVIDIA GeForce RTX 4090 GPUs. For the LoveDA dataset, due to the out-of-memory problem, all the experiments are run on a server with two NVIDIA A40 GPUs. We set the batch size to 8 for all the methods. The weak data augmentations include random resize, flip, and rotation. The strong data augmentations are CutMix [59] and color jitter, and the adopted feature perturbation is dropout, which is the same as in UniMatch [36].

TABLE II
RESULTS OF DIFFERENT METHODS ON THE WHDLD DATASET IN TERMS OF mIoU (%) UNDER DIFFERENT DATA SPLITS

Methods	1/24	1/16	1/8	1/4
Onlysup	53.6	55.2	58.0	60.5
FixMatch [35] [NeurIPS'20]	56.0	57.6	59.8	60.8
U2PL [27] [CVPR'22]	57.1	58.3	59.9	61.1
WSCL [37] [TGRS'23]	56.8	58.6	59.8	61.1
CCVC [63] [CVPR'23]	56.4	57.8	60.1	61.2
UniMatch [36] [CVPR'23]	57.4	58.8	60.4	61.5
CorrMatch [39] [CVPR'24]	56.7	57.7	59.4	60.7
CrossMatch	**58.7**	**60.3**	**61.5**	**62.0**

E. Experimental Results

1) Results on WHDLD: We show the results of different methods under different data splits for training and testing in Table II, where the "1/24" means 1/24 of the labeled data is used for training. The results show that in general, our method CrossMatch yields the best performance in terms of mIoU under different data splits. Compared with the supervised baseline Onlysup, all semi-supervised methods yield better results, demonstrating the benefit of incorporating

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

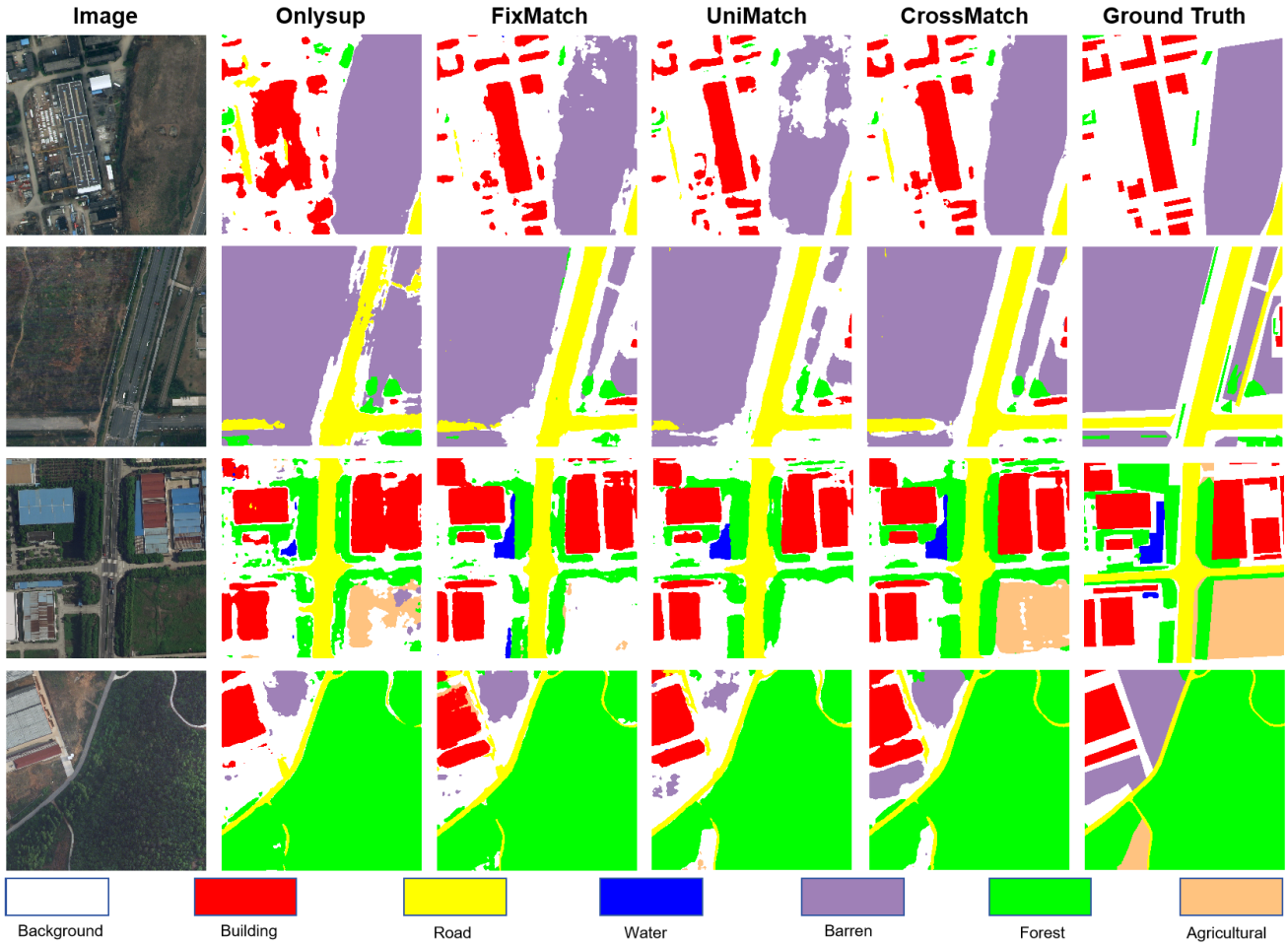


Fig. 6. Visual results of our proposed method compare with only-supervised baseline and other advanced semi-supervised methods on the LoveDA dataset.

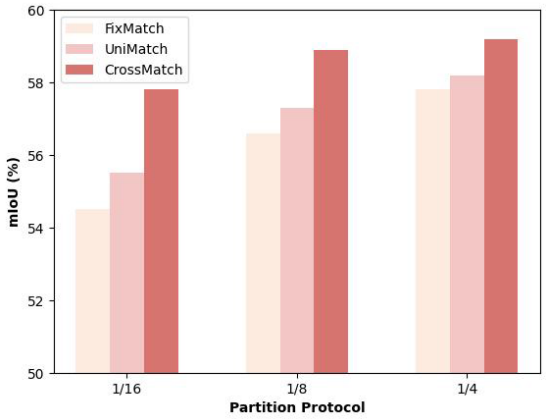


Fig. 7. Pseudo-label quality comparison between different methods by using the mIoU, which is computed by using the pseudo-labels of weakly augmented unlabeled images and the corresponding ground truth.

TABLE III
RESULTS OF DIFFERENT METHODS ON THE POTSDAM DATASET IN TERMS OF mIoU (%) UNDER DIFFERENT DATA SPLITS

Methods	1/32	1/16	1/8	1/4
Onlysup	61.5	64.9	70.2	73.8
FixMatch [35] [NeurIPS'20]	69.4	71.5	73.5	75.7
U2PL [27] [CVPR'22]	67.2	71.3	73.8	75.9
WSCL [37] [TGRS'23]	69.9	72.3	73.9	75.4
CCVC [63] [CVPR'23]	69.6	72.4	74.7	76.2
UniMatch [36] [CVPR'23]	70.7	72.6	74.8	**76.4**
CorrMatch [39][CVPR'24]	70.5	71.0	73.6	75.9
CrossMatch	**71.9**	**73.8**	**75.5**	**76.4**

unlabeled information. Our method outperforms the supervised baseline by 5.1%, 5.1%, 3.5%, and 1.5% under the 1/24, 1/16, 1/8, and 1/4 data splits, respectively, which indicates a larger performance gap between the supervised model and the semi-supervised model when fewer labeled data are used. Compared with all other SSSS methods, CrossMatch yields better performance under different settings. Our method CrossMatch obtains improved mIoU by 2.7%, 2.0%, 1.7%, 2.2%, 1.5%, and 2.6% compared with FixMatch [35], U2PL [27], WSCL [37], CCVC [63], UniMatch [36], and CorrMatch [39] for the case with 1/16 data split, respectively. Compared with the single-view weak-to-strong methods FixMatch, WSCL, UniMatch, and CorrMatch, our CrossMatch consistently yields improved performance, demonstrating the effectiveness of our cross-view learning strategy. Fig. 4 shows the visual segmentation results of different methods on the WHDLD dataset. The results indicate that the segmentation map of our method

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF DIFFERENT METHODS ON THE LOVEDA DATASET UNDER DIFFERENT DATA SPLITS. THE mIoU (%) AND CLASS-SPECIFIC IoUs (%) ARE REPORTED

Methods	Background (35.9%)	Building (9.5%)	Road (5.0%)	Water (8.5%)	Barren (4.9%)	Forest (12.6%)	Agricultural (23.6%)	mIoU
1/40 labeled images								
Onlysup	48.0	37.5	36.1	55.3	27.3	53.2	57.1	45.9
FixMatch [35] [NeurIPS'20]	50.3	52.1	53.3	69.9	28.7	56.4	62.3	53.3
U2PL [27] [CVPR'22]	47.4	47.5	50.9	69.8	30.0	56.5	63.8	52.3
WSCL [37] [TGRS'23]	52.9	52.1	52.8	68.1	31.2	57.7	63.8	54.1
CCVC [63] [CVPR'23]	**54.5**	57.1	56.4	70.7	29.3	57.4	65.6	55.8
UniMatch [36] [CVPR'23]	52.8	55.9	54.6	70.1	30.2	**59.3**	65.5	55.5
CorrMatch [39] [CVPR'24]	52.7	56.8	54.6	70.5	30.1	57.7	64.1	55.2
CrossMatch	52.4	**59.5**	**59.0**	**72.9**	**38.0**	58.7	**67.1**	**58.2**
1/16 labeled images								
Onlysup	50.8	51.0	49.5	65.9	33.5	57.1	63.6	53.1
FixMatch [35] [NeurIPS'20]	55.5	58.4	55.4	72.2	37.4	59.3	66.9	57.9
U2PL [27] [CVPR'22]	53.5	58.3	56.6	72.6	38.0	59.9	67.3	58.0
WSCL [37] [TGRS'23]	53.8	55.8	55.2	71.6	37.5	61.3	66.6	57.4
CCVC [63] [CVPR'23]	**55.9**	**61.5**	58.6	73.5	37.5	61.6	66.9	59.3
UniMatch [36] [CVPR'23]	55.7	59.2	57.0	72.4	37.7	60.9	67.6	58.6
CorrMatch [39] [CVPR'24]	54.6	59.0	55.3	71.9	38.3	60.5	66.8	58.1
CrossMatch	55.7	61.4	**60.3**	**73.9**	**40.1**	**62.5**	**68.2**	**60.3**
1/8 labeled images								
Onlysup	52.8	56.4	53.4	69.8	38.0	59.7	66.2	56.6
FixMatch [35] [NeurIPS'20]	55.9	60.9	58.8	73.1	40.6	62.3	69.2	60.1
U2PL [27] [CVPR'22]	54.8	60.9	59.4	72.0	40.3	61.6	68.4	59.6
WSCL [37] [TGRS'23]	56.6	60.0	58.5	72.9	38.6	61.1	68.0	59.4
CCVC [63] [CVPR'23]	55.7	61.8	60.8	74.5	40.8	62.6	69.5	60.8
UniMatch [36] [CVPR'23]	56.8	61.1	59.6	73.3	39.4	61.7	68.3	60.0
CorrMatch [39] [CVPR'24]	**57.2**	60.6	58.9	73.4	40.6	61.7	69.0	60.2
CrossMatch	56.3	**62.6**	**61.4**	**75.1**	**41.8**	**63.7**	**70.0**	**61.5**
1/4 labeled images								
Onlysup	55.7	58.8	57.9	72.8	42.3	63.2	68.5	58.9
FixMatch [35] [NeurIPS'20]	56.9	61.9	60.6	73.8	43.3	64.2	69.8	61.5
U2PL [27] [CVPR'22]	57.4	61.5	60.6	74.6	42.9	64.3	69.2	61.5
WSCL [37] [TGRS'23]	57.6	61.3	59.3	74.3	43.4	63.2	68.6	61.1
CCVC [63] [CVPR'23]	57.4	62.1	**60.9**	74.7	44.0	64.1	70.0	61.9
UniMatch [36] [CVPR'23]	58.0	61.1	60.8	74.2	**44.6**	63.5	69.8	61.7
CorrMatch [39] [CVPR'24]	**58.3**	61.8	60.8	**74.9**	43.1	64.6	**70.4**	62.0
CrossMatch	57.2	**62.6**	60.4	74.7	**44.6**	**64.8**	70.2	**62.1**

matches the best to the ground truth. It is observed that both the supervised baseline Onlysup and other methods [35], [36] based on single-view weak-to-strong consistency methods struggle to accurately identify certain land-cover objects, such as buildings and roads, particularly when these objects exhibit large intraclass variations. These methods often misclassify the areas with complex structures or diverse appearances. In contrast, our cross-view approach exhibits a powerful ability to recognize these challenging objects, which results in more precise segmentation.

2) Results on Potsdam: We show the results of different methods on the Potsdam dataset in Table III under the data splits of 1/32, 1/16, 1/8, and 1/4. Our method consistently outperforms other SOTA SSSS methods and the supervised baseline Onlysup. A similar conclusion to the results in the WHDLD dataset can be observed that the SSSS methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

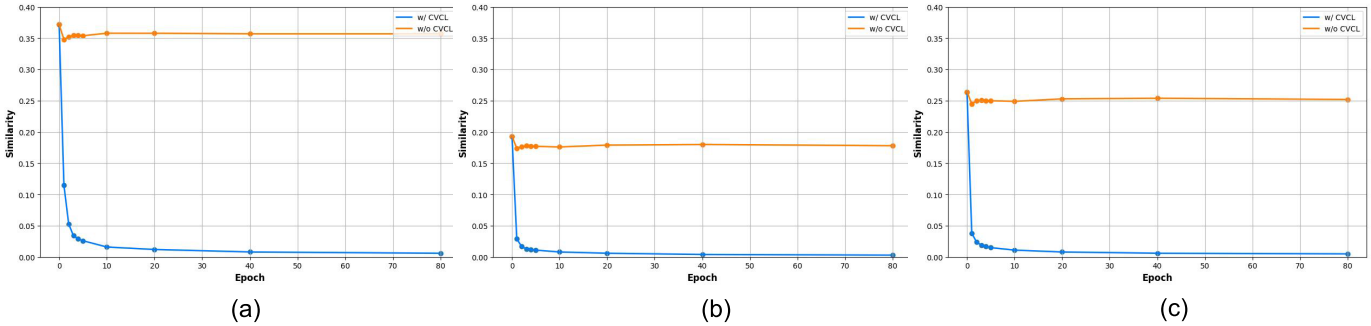


Fig. 8. Cosine similarity of the features between two views. w/o CVCL: the contrastive loss is not used; $w/$ CVCL: the contrastive loss is used. (a) Similarity of the features $f_{i,n}^{\alpha}$ and $f_{3-i,n}^{\alpha}$ between two views. (b) Similarity of the perturbed features $f_{i,n}^{\alpha,fp}$ and $f_{3-i,n}^{\alpha,fp}$ between different views. (c) Similarity of the features $f_{i,n}^{\alpha}$ and the perturbed features $f_{3-i,n}^{\alpha,fp}$ between different views.

TABLE V
ABLATION STUDY ON THE WHDLD AND POTSDAM DATASETS UNDER THE 1/16 DATA SPLIT

Schemes	DB	CVWSC	CVCL	VDC	WHDLD	Potsdam
Onlysup					55.2	64.9
1	✓				57.6	68.7
2	✓	✓			59.3	73.2
3	✓	✓	✓		60.0	73.6
4	✓	✓		✓	59.7	73.6
CrossMatch	✓	✓	✓	✓	**60.3**	**73.8**

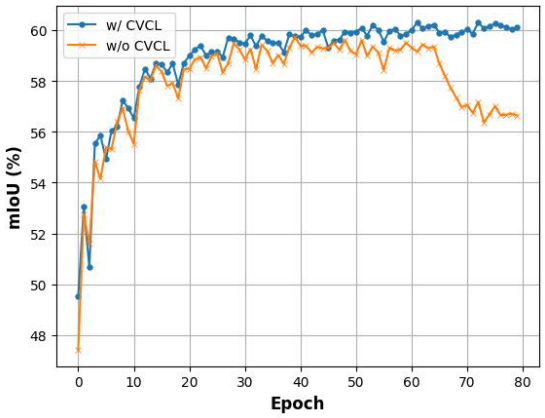


Fig. 9. Influence of the cross-view contrastive loss on the overall mIoU during the model training.

yield larger performance improvement when the labeled data are scarce. UniMatch obtains the second-best performance, yielding improvement over FixMatch [35] by 1.3%, 1.1%, 1.3%, and 0.7% under different data splits. This can be mainly attributed to the enhanced perturbation space. Our method, in contrast to UniMatch [36], revisits the challenges in remote sensing images with more focus on improving the quality of pseudo-labels. It is observed that CrossMatch outperforms the SOTA UniMatch by 1.2%, 1.2%, 0.7% under the 1/32, 1/16, 1/8 partition protocols and obtains a comparable result under the 1/4 partition protocol. We show the visual results of different methods in Fig. 5. It is clear that the segmentation map produced by our method shows better performance in capturing fine-grained boundary details and is more accurate than other methods, especially for the building and clutter classes. While other methods often produce fuzzy segmentations that fail to clearly define boundaries.

3) Results on LoveDA: We show the overall performance and the class-specific performance of different methods on a more complex LovdDA dataset in Table IV. It is observed that our method performs the best in terms of mIoU under different data splits. Compared to the second-best method, our method consistently outperforms by 2.4%, 1.0%, 0.7%, and 0.1% on the 1/40, 1/16, 1/8, and 1/4 splits, respectively. For the class-specific IoU, our method outperforms others in most cases. We notice that our model yields more significant improvement on the minor classes "barren" and "road" compared with the major classes especially when fewer labeled data are utilized. For instance, CrossMatch obtains IoU improvements over FixMatch [35] by 9.3% for the barren and by 5.5% for the road when 1/40 labeled images are used for training, which are much higher than that of the major class background where the improvement is 2.1%. This demonstrates that our method is more effective in dealing with the data with imbalanced class distribution. In addition, compared with the semi-supervised methods, the performance degradation is more prominent for supervised methods when fewer labeled data are used. We show the visual results in Fig. 6. It is observed that CrossMatch obtains better segmentation performance than others for the barren, road, and building classes. The prediction of UniMatch [36] is often biased to the major class background and there are more erroneous segmentation results for the minor classes.

F. Ablation Studies

1) Influence of Different Components: We investigate the influence of different components on the performance of our model on the WHDLD and Potsdam datasets under the 1/16 data split. We show the results of different reduced versions of our model in Table V. DB represents the strategy of using dual branches for feature extraction. CVWSC means that the weak-to-strong consistency scheme is adopted in each branch. CVCL is the cross-view contrastive loss. VDC is the view-discrepancy constraint. The results of the supervised baseline Onlysup are also reported. In general, the different versions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

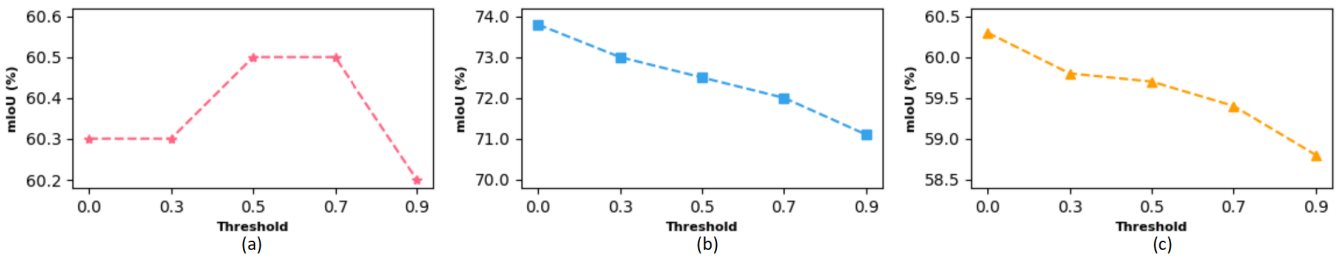


Fig. 10. Performance analysis with different pseudo-labels confidence thresholds under 1/16 split on three datasets. (a) WHDLD, (b) Potsdam, and (c) LoveDA.

of our model outperform the supervised baseline Onlysup. The results in Table V verify the effectiveness of different components in our model. Specifically, the cross-view weak-to-strong consistency strategy leads to mIoU improvements by 1.7% on WHDLD and by 4.5% on Potsdam. The cross-view contrastive loss and view-discrepancy constraint improves the performance of Scheme 2 by 0.7%, 0.4%, and 0.4%, 0.4% on the two datasets, respectively. Compared with Onlysup, CrossMatch obtains improved mIoUs of 5.1% and 8.9% on the two datasets.

2) Investigation of the Pseudo-Labels: We compare the quality of pseudo-labels of different methods here. We first obtain the resulting pseudo-labels for different methods by transforming the related soft labels of unlabeled data to the hard labels by using the argmax operator. For CrossMatch, we obtain the pseudo-labels of the nth image by $\hat{y}_{i,n}^{w} = \text{argmax}(p_{i,n}^{w})$. Then, we calculate the mIoU based on the obtained pseudo-labels and ground truth. Similar operations are applied in FixMatch and UniMatch. We show the results in Fig. 7. It is observed that the generated pseudo-labels by CrossMatch consistently yield the highest mIoU under different partition protocols, demonstrating that our model can yield more reliable pseudo-labels with unlabeled data.

3) The Influence of the Cross-View Contrastive Loss: The cross-view contrastive loss promotes the two views to capture distinct features of the input image, which is important in our model. We show its influence on the feature extraction of the two views in Fig. 8. The results indicate that without imposing the cross-view contrastive loss, the similarities of the features between the two views are much higher than that using the loss. This demonstrates the effectiveness of our cross-view contrastive loss. Furthermore, we show its influence on the model training in Fig. 9. It is observed that when the model learns identical features without the loss and causes the network to collapse, the performance is significantly degraded. In contrast, with the loss, the performance of our model is gradually increased and gets saturated in the end.

3) Influence of the View-Discrepancy Constraint: Our view-discrepancy constraint is imposed only on the labeled data. A natural question is that is it better to apply the constraint on the unlabeled data or the whole data. To verify our design, we conduct experiments for both cases on the WHDLD and Potsdam datasets. We show the results in Table VI, where "Sup," "Unsup," and "Sup&Unsup" mean that the view-discrepancy constraint is imposed on the labeled data, unlabeled data, and the whole data, respectively. It is observed that "Sup" yields the best results on the two datasets. The degraded performance of "Unsup" and "Sup&Unsup" can be mainly attributed to the false supervision that is introduced by the erroneous pseudo-labels of the unlabeled data.

TABLE VI
INFLUENCE OF THE VIEW-DISCREPANCY CONSTRAINT

Sup	Unsup	Sup&Unsup	WHDLD	Potsdam
✓			**58.7**	**73.8**
	✓		58.2	73.5
		✓	58.5	73.7

TABLE VII
COMPARISON BETWEEN DIFFERENT METHODS IN TERMS OF COMPUTATIONAL COST

Method	Parameters	Memory	Training Speed	mIoU
FixMatch [35]	**40.5 M**	**10.1 G**	**1.3 min/epoch**	57.6
UniMatch [36]	**40.5 M**	14.2 G	1.5 min/epoch	58.8
CorrMatch [39]	45.2 M	15.9 G	2.8 min/epoch	57.7
CrossMatch	80.9 M	18.5 G	2.8 min/epoch	**60.3**

4) Influence of Confidence Threshold: The confidence threshold can be carefully tuned to reduce the number of noisy pseudo-labels. In our article, we set the confidence threshold as zero for all experiments of CrossMatch. In order to investigate the influence of the confidence threshold on the performance of our method, we set the threshold in the range of 0, 0.3, 0.5, 0.7, and 0.9 and report the results in Fig. 10. We can see that a better performance can be achieved by setting a suitable value on WHDLD dataset. In the other two datasets, we observe degraded performance when using a larger threshold. This might be caused by the fact that most pseudo-labels are considered unreliable and therefore discarded.

5) Comparison of Computational Cost: We also analyze the computational cost of different SSSS methods and our CrossMatch training under 1/16 protocol on the WHDLD dataset. As shown in Table VII, it can be seen that FixMatch [35] has the lowest cost, but its performance is relatively poor, with a segmentation result of only 57.6% mIoU. The training time of our method is close to CorrMatch [39], while the performance has significantly improved by 2.6%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

Compared to these methods based on single-view weak-to-strong consistency, our method adopts two independent neural networks for feature extraction and therefore shows a higher computational burden. However, the performance gain is significant in terms of mIoU. With the 1/16 split, we achieve a superior result that is competitive with other methods under the 1/8 split.

V. Conclusion

In this article, we develop a novel cross-view matching-based method for remote sensing image segmentation, which effectively utilizes the labeled and unlabeled data. The proposed method excavates complementary information from different views by using unshared subnets and performs information exchange of the two views by using cross-view supervision with the help of weak-to-strong consistency strategy. To achieve this, a cross-view weak-to-strong consistency loss, a cross-view contrastive loss, and a cross-view discrepancy loss are specially designed. Extensive results demonstrate the effectiveness of our design and the superior performance of CrossMatch over the state-of-the-art especially when fewer labeled data are available.

References

[1] S. Huang, H. Zeng, H. Chen, and H. Zhang, "Spatial and cluster structural prior guided subspace clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511115.

[2] Z. Li, H. Zhang, F. Lu, R. Xue, G. Yang, and L. Zhang, "Breaking the resolution barrier: A low-to-high network for large-scale high-resolution land-cover mapping using low-resolution labels," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 244–267, Oct. 2022.

[3] J. Sun, W. He, and H. Zhang, "G2LDIE: Global-to-local dynamic information enhancement framework for weakly supervised building extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5406714.

[4] J. Li, W. He, W. Cao, L. Zhang, and H. Zhang, "UANet: An uncertainty-aware network for building extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608513.

[5] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[6] H. Chen, H. Zhang, G. Yang, S. Li, and L. Zhang, "A mutual information domain adaptation network for remotely sensed semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537316.

[7] L. Wang, B. Zuo, Y. Le, Y. Chen, and J. Li, "Penetrating remote sensing: Next-generation remote sensing for transparent Earth," *Innov.*, vol. 4, no. 6, Nov. 2023, Art. no. 100519.

[8] X. Xu et al., "A novel object detection method for solid waste incorporating a weighted deformable convolution," *Photogrammetric Eng. Remote Sens.*, vol. 89, no. 11, pp. 679–689, Nov. 2023.

[9] Y.-P. Zhao, H. Li, Y. Chen, Z. Wang, and X. Li, "Hyperspectral anomaly detection via structured sparsity plus enhanced low-rankness," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515115.

[10] X. Xu et al., "A data augmentation strategy combining a modified pix2pix model and the copy-paste operator for solid waste detection with remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8484–8491, 2022.

[11] X. Ma et al., "STNet: Spatial and temporal feature fusion network for change detection in remote sensing images," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 2195–2200.

[12] T. Luo, M. Du, J. Shi, X. Chen, B. Zhao, and S. Huang, "Contextuality helps representation learning for generalized category discovery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2024, pp. 687–693.

[13] H. Qin, Q. Shen, H. Zeng, Y. Chen, and G. Lu, "Generalized nonconvex low-rank tensor representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526612.

[14] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524017.

[15] X. Ma et al., "SACANet: Scene-aware class attention network for semantic segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 828–833.

[16] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 6795–6799.

[17] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[18] S. Huang, H. Zhang, and A. Pižurica, "A structural subspace clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509515.

[19] X. Ma et al., "LoG-CAN: Local-global class-aware network for semantic segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[20] R. Guan et al., "Spatial–spectral graph contrastive clustering with hard sample mining for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535216.

[21] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.

[22] X. Ma et al., "DOCNet: Dual-domain optimized class-aware network for remote sensing image segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[23] R. Guan et al., "Contrastive multiview subspace clustering of hyperspectral images based on graph convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510514.

[24] S. Huang, H. Zhang, and A. Pižurica, "Hybrid-hypergraph regularized multiview subspace clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505816.

[25] H. Hu, F. Wei, H. Hu, Q. Ye, J. Cui, and L. Wang, "Semi-supervised semantic segmentation via adaptive equalization learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 22106–22118.

[26] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-training work better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4268–4277.

[27] Y. Wang et al., "Semi-supervised semantic segmentation using unreliable pseudo-labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4248–4257.

[28] D. Guan, J. Huang, A. Xiao, and S. Lu, "Unbiased subclass regularization for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9968–9978.

[29] Y. Jin, J. Wang, and D. Lin, "Semi-supervised semantic segmentation via gentle teaching assistant," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 2803–2816.

[30] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.

[31] Z. Ke, D. Qiu, K. Li, Q. Yan, and R. W. Lau, "Guided collaborative training for pixel-wise semi-supervised learning," in *Proc. 16th Eur. Conf. Comput. Vis.*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 429–445.

[32] G. French, T. Aila, S. Laine, M. Mackiewicz, and G. D. Finlayson, "Semi-supervised semantic segmentation needs strong, high-dimensional perturbations," in *Proc. ICLR*, 2019, p. 1914. [Online]. Available: https://api.semanticscholar.org/CorpusID:202889021

[33] V. Olsson, W. Tranheden, J. Pinto, and L. Svensson, "ClassMix: Segmentation-based data augmentation for semi-supervised learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jan. 2021, pp. 1369–1378.

[34] Y. Liu, Y. Tian, Y. Chen, F. Liu, V. Belagiannis, and G. Carneiro, "Perturbed and strict mean teachers for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4258–4267.

[35] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.

[36] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.

[37] X. Lu et al., "Weak-to-strong consistency learning for semisupervised image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510715.

[38] W. Huang, Y. Shi, Z. Xiong, and X. X. Zhu, "AdaptMatch: Adaptive matching for semisupervised binary segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625416.

[39] B. Sun, Y. Yang, L. Zhang, M.-M. Cheng, and Q. Hou, "CorrMatch: Label propagation via correlation matching for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3097–3107.

[40] S. Huang, H. Zhang, J. Xue, and A. Pižurica, "Heterogeneous regularization-based tensor subspace clustering for hyperspectral band selection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9259–9273, Mar. 2023.

[41] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Munich, Germany. Cham, Switzerland: Springer, 2015, pp. 234–241.

[42] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[43] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[44] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2020.

[45] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[46] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.

[47] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[48] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.

[49] J. Jain, J. Li, M. Chiu, A. Hassani, N. Orlov, and H. Shi, "OneFormer: One transformer to rule universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2989–2998.

[50] R. Ke, A. I. Aviles-Rivero, S. Pandey, S. Reddy, and C.-B. Schönlieb, "A three-stage self-training framework for semi-supervised semantic segmentation," *IEEE Trans. Image Process.*, vol. 31, pp. 1805–1815, 2022.

[51] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–10.

[52] X. Lai et al., "Semi-supervised semantic segmentation with directional context-aware consistency," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1205–1214.

[53] H. Xu, L. Liu, Q. Bian, and Z. Yang, "Semi-supervised semantic segmentation with prototype-based consistency regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 26007–26020.

[54] S. Qiao, W. Shen, Z. Zhang, B. Wang, and A. Yuille, "Deep co-training for semi-supervised image recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 135–152.

[55] D.-D. Chen, W. Wang, W. Gao, and Z.-H. Zhou, "Tri-Net for semi-supervised deep learning," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 2014–2020.

[56] L. Yang et al., "Deep co-training with task decomposition for semi-supervised domain adaptation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8906–8916.

[57] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.

[58] J. Fan, B. Gao, H. Jin, and L. Jiang, "UCC: Uncertainty guided cross-head cotraining for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9937–9946.

[59] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6023–6032.

[60] Z. Shao, W. Zhou, X. Deng, M. Zhang, and Q. Cheng, "Multilabel remote sensing image retrieval based on fully convolutional network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 318–328, 2020.

[61] F. Rottensteiner et al., "THE ISPRS benchmark on urban object classification and 3D building reconstruction," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. I-3, pp. 293–298, Jul. 2012.

[62] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[63] Z. Wang, Z. Zhao, X. Xing, D. Xu, X. Kong, and L. Zhou, "Conflict-based cross-view consistency for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19585–19595.

[64] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



Ruizhong Liu received the B.S. degree in computer science and technology from the School of Computer Science, China University of Geosciences, Wuhan, China, in 2024.

He is currently a Research Assistant with the School of Computer Science, China University of Geosciences. His research interests include computer vision, semi-supervised learning, and remote sensing image interpretation.



Tingzhang Luo is currently pursuing the B.S. degree in computer science and technology from the School of Computer Science, China University of Geosciences, Wuhan, China.

His research interests include developing the label-efficient methodologies for classification task in computer vision and image processing.



Shaoguang Huang (Member, IEEE) received the M.S. degree in telecommunication and information system from Shandong University, Jinan, China, in 2015, and the Ph.D. degree in computer science engineering from Ghent University, Ghent, Belgium, in 2019.

He is currently a Full Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. His area of interests includes image processing, sparse representation, clustering, hyperspectral image analysis, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.



Yuwei Wu received the bachelor's degree in geographic information system from Wuhan University, Wuhan, China, in 2005, and the master's degree from Land Resources and Utilization, Jiangxi Agricultural University, Nanchang, China, in 2011.

He has been working with Jiangxi Provincial Institute of Land Spatial Survey and Planning, Nanchang, since 2006. He is currently the Director of the Survey and Evaluation Institute of the Cultivated Land Protection Research Center. He has undertaken and completed more than 40 national and provincial key projects, and published more than 20 articles. His main research direction is cultivated land protection.



Zhen Jiang received the bachelor's degree in surveying and mapping engineering from Heilongjiang Institute of Technology, Harbin, China, in 2005, and the master's degree in computer technology engineering from Wuhan University, Wuhan, China, in 2010.

Since 2005, he has been working as a Senior Engineer with Jiangxi Provincial Institute of Land Spatial Survey and Planning, Nanchang, China, Certified Urban-Rural Planner, and Registered Surveyor. He has undertaken more than 30 national and provincial key projects and published more than ten articles. His main research direction is surveying geographic information and land spatial planning.



Hongyan Zhang (Senior Member, IEEE) received the B.S. degree in geographic information systems and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2005 and 2010, respectively.

From 2010 to 2022, he was a Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University. Currently, he is the Dean of the School of Computer Science, China University of Geosciences, Wuhan. He is a Young Chang-Jiang Scholar appointed by the Ministry of Education of China. He has authored/co-authored more than 100 research articles. His research interests include high-dimensional data intelligent processing and agricultural remote sensing.

Dr. Zhang scored first in the Data Fusion Contest of 2019 and 2021 organized by the IEEE Image Analysis and Data Fusion Technical Committee. He serves as an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, *Photogrammetric Engineering and Remote Sensing and Computers and Geosciences*. He is a reviewer for more than 40 international academic journals, including *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE TRANSACTIONS ON IMAGE PROCESSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:30:09 UTC from IEEE Xplore. Restrictions apply.